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import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.svm import SVC
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, confusion_matrix

df=pd.read_csv("car_acceptability.csv")
df

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	buying price	maintenance cost	number of doors	number of persons
0	vhigh	vhigh	2	2
1	vhigh	vhigh	2	2
2	vhigh	vhigh	2	2
3	vhigh	vhigh	2	2
4	vhigh	vhigh	2	2
...	...	...	...	...
1722	low	low	5	5
1723	low	low	5	5
1724	low	low	5	5
1725	low	low	5	5
1726	low	low	5	5

	lug_boot	safety	evaluation
0	small	med	unacc
1	small	high	unacc
2	med	low	unacc
3	med	med	unacc
4	med	high	unacc
...	...	...	...
1722	med	med	good
1723	med	high	vgood
1724	big	low	unacc
1725	big	med	good
1726	big	high	vgood

[1727 rows x 7 columns]

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le = LabelEncoder()

df["buying price"]=le.fit_transform(df["buying price"])
df["maintenance cost"]=le.fit_transform(df["maintenance cost"])
df["lug_boot"]=le.fit_transform(df["lug_boot"])
df["safety"]=le.fit_transform(df["safety"])
df["evaluation"]=le.fit_transform(df["evaluation"])

df

```

	buying price	maintenance cost	number of doors	number of persons \
0	3	3	2	
2				
1	3	3	2	
2				
2	3	3	2	
2				
3	3	3	2	
2				
4	3	3	2	
2				
...	...	...	...	...
..				
1722	1	1	5	
5				
1723	1	1	5	
5				
1724	1	1	5	
5				
1725	1	1	5	
5				
1726	1	1	5	
5				
	lug_boot	safety	evaluation	
0	2	2	2	
1	2	0	2	
2	1	1	2	
3	1	2	2	
4	1	0	2	
...	...	...	...	
1722	1	2	1	
1723	1	0	3	
1724	0	1	2	
1725	0	2	1	
1726	0	0	3	

[1727 rows x 7 columns]

```

X= df.drop("evaluation",axis=1)
Y=df["evaluation"]

X_train,X_test,Y_train,Y_test=train_test_split(X,Y,test_size=0.2,random_state=42)

svm_model=SVC(kernel="linear")
svm_model.fit(X_train,Y_train)

SVC(kernel='linear')

Y_pred=svm_model.predict(X_test)

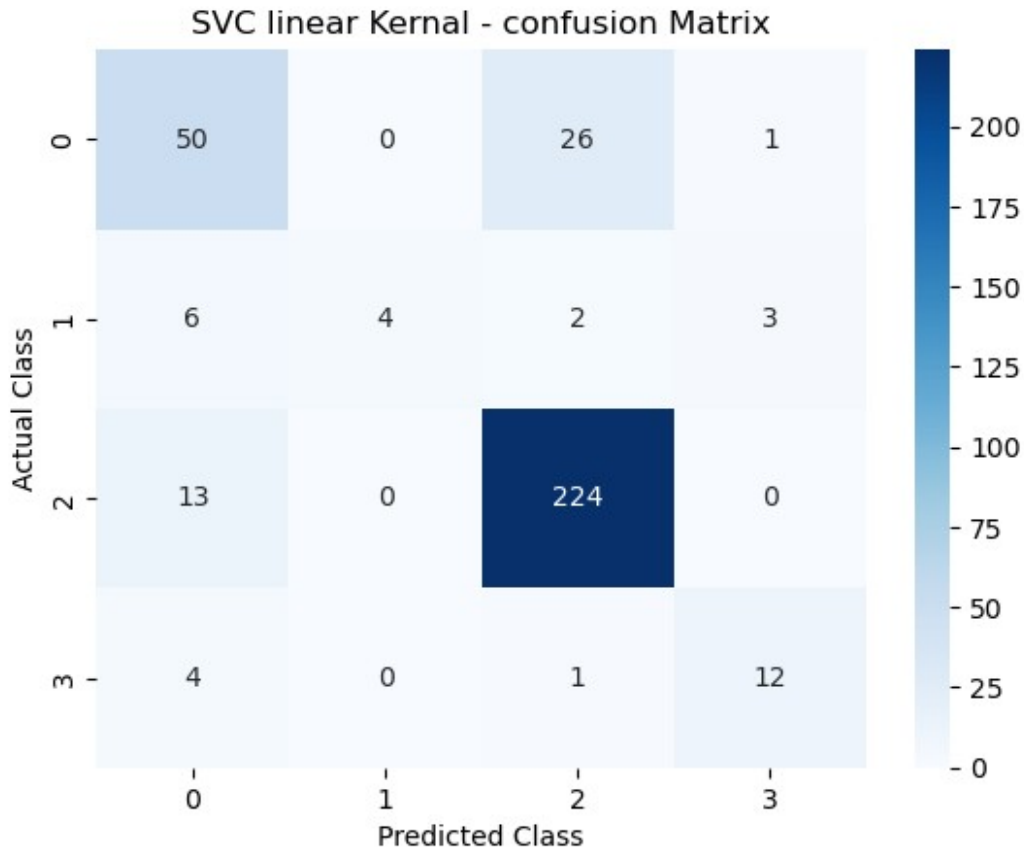
print("Accuracy:",accuracy_score(Y_test,Y_pred))
cm=confusion_matrix(Y_test,Y_pred)
print("\nConfusion Matrix:\n",confusion_matrix(Y_test,Y_pred))

Accuracy: 0.7138728323699421

Confusion Matrix:
[[ 12   0  65   0]
 [  0   0  15   0]
 [  2   0 235   0]
 [ 12   0   5   0]]

sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("SVC linear Kernel - confusion Matrix")
plt.show()

```



```
svm_model=SVC(kernel="poly")
svm_model.fit(X_train,Y_train)

SVC(kernel='poly')

Y_pred=svm_model.predict(X_test)

print("Accuracy:",accuracy_score(Y_test,Y_pred))
cm=confusion_matrix(Y_test,Y_pred)
print("\nConfusion Matrix:\n",confusion_matrix(Y_test,Y_pred))

Accuracy: 0.838150289017341

Confusion Matrix:
[[ 50   0  26   1]
 [  6   4   2   3]
 [ 13   0 224   0]
 [  4   0   1  12]]

sns.heatmap(cm,annot=True,fmt="d",cmap="Blues")
plt.xlabel("Predicted Class")
plt.ylabel("Actual Class")
plt.title("SVC polynomial Kernal - confusion Matrix")
plt.show()
```

